

Programming SQL Server 2019

Table constraints

Topics

- Data integrity
 - What is data integrity?
 - Various types of data integrity
- Constraints
- Role of database constraints
- Constraints in SQL Server
- Applying constraints to a column
- Applying constraints to an existing table
- Dropping a constraint from a table
- Viewing constraints of a table
- Default constraint
 - Applying default constraints in table creation
 - Applying default constraints after table creation
- Check constraint
 - What is it?
 - Applying check constraints

Homework

A. Create a table 'Orders' with columns as below

Column Name	Data Type	Auto increment	Allow Nulls	Constraint
OrderId	Int	Yes	No	Primary Key
OrderDate	Date		No	Default value Current Date
ProductId	Int		No	
Quantity	Int		No	always over 0
ShipDate	Date		Yes	not before Orderdate

- B. Show me Constraints in the table
C. Drop default constraint from OrderDate column in the table
D. Drop the table

Important points

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Summary

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Data integrity

- Data integrity refers to maintaining and assuring the accuracy and consistency of data
- To enforce data integrity, you create rules at design-time that protect your data from becoming corrupt.

Types of data integrity

- The following categories of the data integrity exist
 - **Entity Integrity** ensures that there are no duplicate rows in a table.
 - **Domain Integrity** enforces valid entries for a given column by restricting the type, the format, or the range of possible values.
 - **Referential integrity** ensures relationships between data rows in multiple tables
 - **User-Defined Integrity** enforces some specific business rules that do not fall into entity, domain, or referential integrity categories.
- Each of these categories of the data integrity can be enforced by the appropriate constraints
- You can also enforce data integrity in procedural way – using stored procedure, triggers etc.

Constraints

- Database constraints enforce data integrity
- Constraints prevent from invalid data entering into the system
- Constraints can be applied to table or columns

Constraints in SQL Server

- Microsoft SQL Server supports the following constraints:
 - PRIMARY KEY
 - UNIQUE KEY
 - FOREIGN KEY
 - CHECK CONSTRAINT
 - DEFAULT CONSTRAINT
- Data type, Nullability are also constraints – you have already used these

Applying constraint to a column

- The basic syntax is:
column-name data-type CONSTRAINT constraint-name constraint-definition
- You can omit constraint-name, like below
column-name data-type constraint-definition

Applying constraint to a table

- You have to use ALTER table statement
- The syntax is
ALTER TABLE table-name
ADD CONSTRAINT constraint-name constraint-definition
- You can omit CONSTRAINT constraint-name portion

Dropping a constraint from a table

- You have to use ALTER table statement
- The syntax is
ALTER TABLE table-name
DROP CONSTRAINT constraint-name

Viewing constraints of a table

- You have to use sp_helpconstraint stored procedure
- Use
EXECUTE sp_helpconstraint 'table-name'

Default constraint (value)

- Specifies a default value for a column if none is given

- If default value for a column is specified and you do not give value for that column while inserting a new row, SQL server uses default value for that column

Applying defaults to columns during table creation

Try this

```
CREATE TABLE trainee
(
    id CHAR(7) NOT NULL,
    name VARCHAR (30) NOT NULL,
    course VARCHAR(20) NOT NULL
    CONSTRAINT DF_course DEFAULT 'NA',
    [round] INT DEFAULT 22
)
GO
```

- For course column we provided constraint name but for round column we did not, SQL Server will generate a unique name for it.
- Let's view the constraints

Try this

```
EXEC sp_helpconstraint 'trainee'
GO
```

- The result is

Object Name
Trainee

constraint_type	constraint_name	..	constraint_keys
DEFAULT on column round	DF__trainee__rou nd__03317E3D	..	((22))
DEFAULT on column course	DF_course	..	('NA')

- Let's insert some rows and see how defaults work

Try this

```
INSERT INTO trainee VALUES ('111601', 'Masud', 'ESAD', 23)
INSERT INTO trainee (id, name, course)
VALUES ('111689', 'Tarek', 'NT')
INSERT INTO trainee (id, name) VALUES ('111543', 'Ikbal')
GO
SELECT * FROM trainee
GO
```

- The result is

id	name	course	round
111601	Masud	ESAD	23
111689	Tarek	NT	22
111543	Ikbal	NA	22

Adding default value for existing column in a table

- You have to do it using ALTER TABLE statement

Try this

```
ALTER TABLE contact ADD DEFAULT 'Unknown' FOR name
GO
EXEC sp_helpconstraint 'contact'
GO
```

- The result is

Object Name
Trainee

constraint_type	constraint_name	..	constraint_keys
DEFAULT on column name	DF__trainee__name__0425A276	..	('Unknown')
DEFAULT on column round	DF__trainee__round__03317E3D	..	((22))
DEFAULT on column course	DF__course	..	('NA')

Check constraint

- Check constraints validates value of a column when a new row is to insert
- It applies check on values, apply rules and rejects value if it does not fit in the rule
- It can limit value by defining:
 - A set / range of values
 - A pattern to match
 - An expression to evaluate

Adding check constraint at column definition

- Syntax is
column-name data-type
CONSTRAINT constraint-name CHECK (Expression)
- CONSTRAINT constraint-name part is optional
 - Expression in check constraint must return either true or false
 - Check constraint can only refer column to the columns of the same table
- Let's try one

Try this

```
CREATE TABLE product
(
    id CHAR (5) NOT NULL
    CHECK (id LIKE 'P-[0-9][0-9][0-9]'),
    name VARCHAR(30) NOT NULL,
    price MONEY NOT NULL CHECK (price > 0)
)
GO
```

- Let's see how CHECK constraints prevents wrong data from entering into the table

Try this

```
INSERT INTO product VALUES ('P-001', 'P1', 99.99)
GO
```

- This one works fine
- But next one fails
INSERT INTO product VALUES ('002', 'P2', 89.99)
GO
- Error message
The INSERT statement conflicted with the CHECK constraint "CK__product__id__08EA5793". The conflict occurred in database "sqlstore", table "dbo.product", column 'id'.
The statement has been terminated.
- This one also fails
INSERT INTO product VALUES ('P-002', 'P2', 0)
GO
- Error message
The INSERT statement conflicted with the CHECK constraint "CK__product__price__09DE7BCC". The conflict occurred in database "sqlstore", table "dbo.product", column 'price'.
The statement has been terminated.

Check constraint based on more than one column

- It must be defined at table level
- Let's try one

Try this

```
CREATE TABLE machine_log
(
    [machine-code] CHAR(3) NOT NULL,
    [acquire-date] DATE NOT NULL,
    [release-date] DATE NOT NULL,
    CHECK ([release-date]> [acquire-date])
)
GO
```

Try this

- This one works
INSERT INTO machine_log
VALUES ('T01', '2013-03-09', '2014-01-12')
GO
- This one fails
INSERT INTO machine_log
VALUES ('G11', '2013-03-09', '2013-03-08')
GO
- Error message
The INSERT statement conflicted with the CHECK constraint "CK__machine_log__0BC6C43E". The conflict occurred in database "sqlstore", table "dbo.machine_log".
The statement has been terminated.

Dropping check constraint

Try this

```
ALTER TABLE machine_log
DROP CONSTRAINT CK__machine_log__0BC6C43E
GO
```

Adding check constraint to an existing table

Try this

```
ALTER TABLE machine_log
ADD CHECK ([release-date]> [acquire-date])
GO
```

- Test it
- The insert will fail again
INSERT INTO machine_log
VALUES ('G11', '2013-03-09', '2013-03-08')
GO
- See data
SELECT * FROM machine_log
GO
- The result

machine-code	acquire-date	release-date
T01	2013-03-09	2014-01-12